

IEM KOLKATA
Advanced Aptitude & Logic
Quantitative Aptitude

Q1) Resultant price of mixture of golden rice is Rs. 480 per kg. This mixture is made from two varieties of golden rice which are having price of Rs. 420 per kg and Rs. 520 per kg. What should be the ratio of quantities of both types of golden rice?

Q2) A mixture consists of some amount of sandalwood oil and 240 litres of water. It is priced at Rs. 275 per litre. Sandalwood oil is priced at Rs. 325 per litre. How much oil is there in the mixture?

Q3) Two containers have mixtures of milk & water, in the ratios 3:2 & 6:5 respectively. In what ratio should the contents be mixed so that the ratio of milk to water in the final mixture is 4:3?

(a) 6:15 (b) 9:14 (c) 10:11 (d) 5:8

Q4) From a container of 50 litres pure milk, 10l milk is taken out & replaced by 10l of water. If this process is repeated thrice, what is the ratio of water to milk finally?

(a) 7:16 (b) 9:16 (c) 61:64 (d) 64:61

Q5) In a mixture, milk & water are in the ratio of 1:4. Some milk is added to the mixture, because of which ratio of milk & water becomes 2:1. How much milk was added as a percentage of the initial mixture?

(a) 400% (b) 200% (c) 240% (d) 140%

Q6) The mean marks of 180 students is 52. If the mean of passed students is 59 and that of failed students is 23, find the number of passed and failed students.

Q7) The average salary of all workers in a factory is Rs. 14000. The average salary of all 80 skilled workers is Rs. 15000 & the average salary of the remaining workers is Rs. 12000. Find the total number of workers in the factory?

(a) 140 (b) 150 (c) 110 (d) 120

Q8) The average weight of 10 workers travelling in a boat is increased by 2.8kg, when one of the workers, who weighs 58kg, is replaced by a new worker. Find the weight of the new worker.

(a) 86kg (b) 84 kg (c) 82 kg (d) 68 kg

Q9) The ratio of ages of A:B is 3:5. If B is 8 years older than A, then what is A's age (in years)?

(a) 12 (b) 16 (c) 20 (d) 18

Q10) 3 years ago, the average age of A & B was 32. Today average age of A, B & C is 30. What is the age of C?

(a) 5 (b) 10 (c) 15 (d) 20

Q11) At the time of their marriage, a couple's average age was 25 years. 2 years later, their child was born. At the present day, their average age is 24 years. How many years ago did the couple get married?

(a) 20 years (b) 10 years (c) 18 years (d) 8 years

Q12) A boy increases his speed to $\frac{9}{5}$ times of his original speed. By this, he reaches his school 30 minutes before the usual time. How much time does he usually take?

(a) 60 mins (b) 67 mins (c) 1 hr 7 min 5 secs (d) 1 hr 7 min 30 secs

Q13) In a 500m race, ratio of speed of A & B is 3:4. If A has a start of 140m, then by how many metres does A win?

(a) 15 (b) 20 (c) 30 (d) 25

Q14) A train covers a certain distance in 45 mins. If its speed is reduced

by 5km/hr, it takes 3 minutes more to cover the same distance. The distance (in km) is:

- (a) 64 (b) 60 (c) 54 (d) 80

Q15) Anita travelled a distance for 11 minutes at the speed of 35km/hr in a auto. She travelled for 11 mins in a taxi at 55 km/hr & finally she travelled for 11 mins by bus at 42 km/hr to reach home. Find her average speed for the whole journey.

- (a) 52 km/hr (b) 64 km/hr (c) 34 km/hr (d) 44 km/hr

Q16) Ramesh sees a thief at a distance of 80m. Ramesh starts chasing the thief who is running at a speed of 5m/s. Ramesh is chasing him with a speed of 7m/s. How much distance does the thief cover before Ramesh catches him?

- (a) 120m (b) 100m (c) 240m (d) 200m

Q17) A travels from Jammu to Delhi and at the same time B starts from Delhi to Jammu. After passing each other, they complete their journeys in 361 hours and 289 hours, respectively. Find the ratio of speed of A to that of B?

- (a) 4:5 (b) 11:12 (c) 17:19 (d) 25:26

Q18) Simran takes twice as long to swim up as to swim down the river and has a speed of 12 km/hr in still water. What is river's speed?

- (a) 2km/hr (b) 3km/hr (c) 4km/hr (d) 6km/hr

Q19) What will be the value of c , if $a(a+b+c) = 85$, $b(a+b+c) = 96$, $c(a+b+c) = 108$?

Q20) What will be the value of $(m+n)$ if we know that $\sqrt{28-6\sqrt{3}} = \sqrt{3}m+n$?

- (a) 1 (b) 2 (c) 3 (d) -1

Q21) Sum of the roots of a quadratic equation is 5 less than the product

of the roots. If one root is 1 more than the other root, find the product of the roots?

- (a) 6 or 3 (b) 12 or 2 (c) 8 or 4 (d) 12 or 4

Q22) How many isosceles triangles with integer sides are possible such that sum of two of the side is 12?

- (a) 11
(b) 6
(c) 17
(d) 23